

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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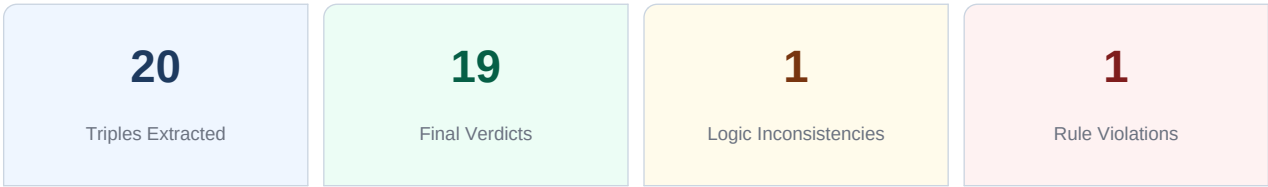
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REPORT OVERVIEW & METHODOLOGY

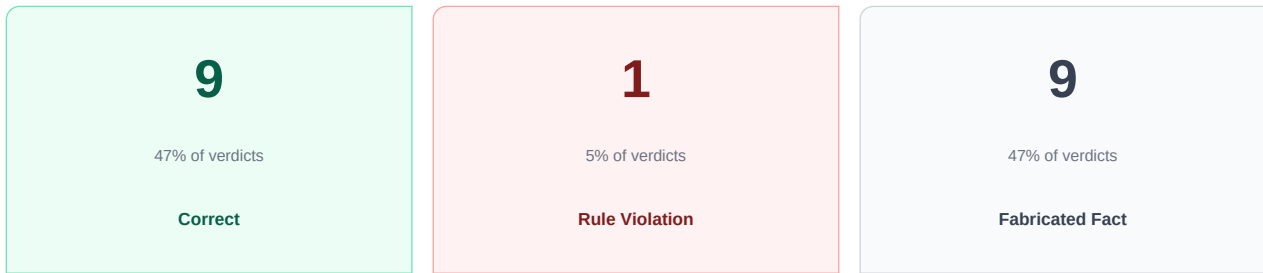
This document outlines the results of the KGVS verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	562	HAS_CREDIT_LIMIT_MULTIPLIER	Not provided		Indicates the multiplier used to determine the credit limit, which is not specified in the provided data for this score.
2	562	HAS_LOAN_TERM	Not provided		Indicates the duration of the loan in months, which is not specified in the provided data for this score.
3	562	HAS_CREDIT_LIMIT_MULTIPLIER	2.07		Indicates the multiplier used to determine the credit limit, as provided in the answer for this score.
4	562	HAS_LOAN_TERM	36 to 48		Indicates the duration range of the loan in months, as provided in the answer for this score.
5	620-660	HAS_RISK_TIER	near prime		Indicates the risk category for credit scores within this range.
6	580-619	HAS_RISK_TIER	subprime		Indicates the risk category for credit scores within this range.
7	580-615	HAS_APPROVAL_ODDS	41%		Indicates the common likelihood of loan approval for all scores in this range.
8	580-615	HAS_MAXIMUM_DEBT_TO_INCOME_RATIO	39%		Indicates the common maximum allowed debt-to-income ratio for all scores in this range.
9	580-615	HAS_LOAN_TERM	36 months		Indicates the common duration of the loan in months for all scores in this range.
10	580-615	HAS_CREDIT_LIMIT_MULTIPLIER	1.61x		Indicates the common credit limit multiplier for all scores in this range.
11	580-615	HAS_DOWN_PAYMENT	12.8%		Indicates the common down payment percentage for all scores in this range.
12	580-615	HAS_DEFAULT_PROBABILITY	11.3%		Indicates the common probability of default for all scores in this range.
13	580	HAS_RISK_TIER	subprime		Indicates the risk category associated with this credit score.
14	580	HAS_INTEREST_RATE	11.91%		Indicates the interest rate associated with this credit score.
15	580	HAS_APPROVAL_ODDS	41%		Indicates the likelihood of loan approval for this credit score.
16	580	HAS_MAXIMUM_DEBT_TO_INCOME_RATIO	39%		Indicates the maximum allowed debt-to-income ratio for this credit score.
17	580	HAS_LOAN_TERM	36 months		Indicates the duration of the loan in months for this credit score.
18	580	HAS_DOWN_PAYMENT	12.8%		Indicates the required down payment percentage for this credit score.
19	580	HAS_CREDIT_LIMIT_MULTIPLIER	1.61x		Indicates the multiplier used to determine the credit limit for this credit score.
20	580	HAS_DEFAULT_PROBABILITY	11.3%		Indicates the probability of default associated with this credit score.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
562	HAS_CREDIT_LIMIT_MULTIPLIER	Not provided	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	MEDIUM
562	HAS_LOAN_TERM	Not provided	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	MEDIUM

Subject	Original Predicate	Object	Canonical Predicate	Conf.
562	HAS_CREDIT_LIMIT_MULTIPLIER	2.07	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	MEDIUM
562	HAS_LOAN_TERM	36 to 48	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	MEDIUM
620-660	HAS_RISK_TIER	near prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
580-619	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
580-615	HAS_APPROVAL_ODDS	41%	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
580-615	HAS_MAXIMUM_DEBT_TO_INCOME_RATIO	39%	MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING	MEDIUM
580-615	HAS_LOAN_TERM	36 months	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	MEDIUM
580-615	HAS_CREDIT_LIMIT_MULTIPLIER	1.61x	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	MEDIUM
580-615	HAS_DOWN_PAYMENT	12.8%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
580-615	HAS_DEFAULT_PROBABILITY	11.3%	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	MEDIUM
580	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
580	HAS_APPROVAL_ODDS	41%	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
580	HAS_MAXIMUM_DEBT_TO_INCOME_RATIO	39%	MAX_DTI_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING	MEDIUM
580	HAS_LOAN_TERM	36 months	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	MEDIUM
580	HAS_DOWN_PAYMENT	12.8%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
580	HAS_CREDIT_LIMIT_MULTIPLIER	1.61x	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	MEDIUM
580	HAS_DEFAULT_PROBABILITY	11.3%	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	MEDIUM

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

Label	Type	Severity	Explanation	Triples Involved
logically_inconsistent	MONOTONICITY_VIOLATION	MEDIUM	Credit score 580 has a lower credit limit multiplier (1.61x) than the lower credit score 562 (2.07), which contradicts the required INCREASING direction for this predicate.	562 -> CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR -> 2.07 580 -> CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR -> 1.61x

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	620-660	RISK_TIER_CLASSIFIED_AS_BORROWER_RANGE_LEVEL	near prime	c_tier_660_739	The credit score range '620-660' includes a score (660) that falls into the 660-739 range, but the risk tier 'near prime' does not contain 'prime'.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
562	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	2.07	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	●●●●●●●● ●●■ 0.90	Object+predicate exist in DB but no matching subject found; subject is fabricated.
580-619	RISK_TIER_CLASSIFIED_AS_BORROWER_RANGE_LEVEL	subprime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	●●●●●●●● ●●■ 0.90	Object+predicate exist in DB but no matching subject found; subject is fabricated.
580-615	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFSET	36 months	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	SUBJECT	●●●●●●●● ●●■ 0.90	Object+predicate exist in DB but no matching subject found; subject is fabricated.
562	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	Not provided	Correct	factually_consistent	—	NONE	—	●●●●●●●● ●●■ 0.90	The object 'Not provided' correctly signals the absence of information for this subject-predicate pair.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
562	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFSET	Not provided	Correct	factually_consistent	—	NONE	—	●●●●●●●● 0.90	The object 'Not provided' correctly signals the absence of information for this subject-predicate pair.
562	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFSET	36 to 48	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● 0.90	The object '36 to 48' asserts a concrete value, but no evidence exists for this subject-predicate pair in the database.
580-615	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	41%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● 0.90	The object '41%' asserts a concrete value, but no evidence exists for this subject-predicate pair in the database.
580-615	MAX_DURATION_ALLOWED_AS_MONTHLY_DEBT_BURDEN_CEILING	39%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● 0.90	The object '39%' asserts a concrete value, but no evidence exists for this subject-predicate pair in the database.
580-615	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	1.61x	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● 0.90	The object '1.61x' asserts a concrete value, but no evidence exists for this subject-predicate pair in the database.
580-615	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	12.8%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● 0.90	The object '12.8%' asserts a concrete value, but no evidence exists for this subject-predicate pair in the database.
580-615	DEFAULT_PROBABILITY_RECORD_AS_HISTORICAL_NONREPAYMENT_RATE	11.3%	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	●●●●●●●● 0.90	The object '11.3%' asserts a concrete value, but no evidence exists for this subject-predicate pair in the database.
580	RISK_TIER_CLASSIFIED_AS_BORROWER_DEFAULT_LEVEL	subprime	Correct	factually_consistent	—	SEMANANTIC	—	●●●●●●●● 0.80	The object 'subprime' semantically matches the DB candidate 'subprime'. <i>Ev: subprime</i>

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	Rule Violation	factually_inconsistent	RULE_VIOLATION	NONE	OBJECT	●●●●●●●● ●●■ 0.90	The object '11.91%' does not strictly match the DB candidate '11.91' for a percentage value. <i>Ev: 11.91</i>
580	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	41%	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●●■ 0.80	The object '41%' semantically matches the DB candidate '41'. <i>Ev: 41</i>
580	MAX_DURATION_PERMITTED_AS_MONTHLY_DEBT_BURDEN_CEILING	39%	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●●■ 0.80	The object '39%' semantically matches the DB candidate '39'. <i>Ev: 39</i>
580	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	36 months	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●●● 1.00	The object '36 months' exactly matches the DB candidate '36 months'. <i>Ev: 36 months</i>
580	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	12.8%	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●●■ 0.80	The object '12.8%' semantically matches the DB candidate '12.8'. <i>Ev: 12.8</i>
580	CREDIT_LIMIT_MULTIPLIER_APPLIED_AS_INCOME_BASED_BORROWING_CEILING_FACTOR	1.61x	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●●■ 0.80	The object '1.61x' semantically matches the DB candidate '1.61'. <i>Ev: 1.61</i>
580	DEFAULT_PROBABILITY_RECORDED_AS_HISTORICAL_NONREPAYMENT_RATE	11.3%	Correct	factually_consistent	—	SEMANTIC	—	●●●●●●●● ●●■ 0.80	The object '11.3%' semantically matches the DB candidate '11.3'. <i>Ev: 11.3</i>

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship that is in db but the subject does not supported by db.